MANAV KULSHRESTHA

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EDUCATION

Ph.D. in Computer Science, Purdue University — Expected 2028

- Specializing in scene understanding and representations for robot decision making and control through machine learning, advised by Prof. Aniket Bera.
- Currently working on vision-language models for robotics manipulation and interactive navigation, and reinforcement learning for behavior modeling in multi-agent systems

B.S. in Computer Science, *Summa Cum Laude* — 2018 - 2022
University of Massachusetts, Amherst

B.S. in Mathematics, *Summa Cum Laude* — 2018 - 2022
University of Massachusetts, Amherst

Relevant Graduate Level Coursework: Machine learning; Robot Learning; Motion Planning; Robotics: Perception Dynamics, and Control; Large Language Models; Advanced Algorithms; Algorithms for Data Science; Advanced Linear Algebra; Regression Analysis...

PATENTS AND SELECTED PUBLICATIONS

- **VLAD-Grasp: Zero-shot Grasp Detection via Vision-Language Models**. IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS). *Manav Kulshrestha*, S. Talha Bukhari, Damon Conover, Aniket Bera.
- **(63/913,578) System and Method for Zero-Shot Multimodal Grasp Synthesis.** Aniket Bera, *Manav Kulshrestha*, S. Talha Bukhari
- **CoReLIN: Constraint-based Reasoning for Zero-shot Lifelong Interactive Navigation**. European Conference on Computer Vision (ECCV). Apoorva Vashisth, *Manav Kulshrestha*, Damon Conover, Guillaume Sartoretti, Aniket Bera.
- **TGrasp6D: Task-Informed Vision-Language Priors and Pose Refinement enable Zero-Shot Task-Oriented 6DoF Grasping.** Under Review at International Conference on Robotics and Automation (ICRA). *Manav Kulshrestha*, Apoorva Vashisth, Damon Conover, Aniket Bera.
- **(18/978,777) Machine Learning Based Selection of Item Manipulation Positions Executed by Robotic Manipulators**. Joshua Hooks, Lillian Ratliff, Bhavana Chandrashekhar, Nicholas Hudson, Hillel Baderman, Amanda Bouman, *Manav Kulshrestha*, Che Wang.
- **Structural Concept Learning via Graph Attention for Multi-Level Rearrangement Planning**. Conference on Robot Learning (CoRL). *Manav Kulshrestha*, Ahmed H. Qureshi.

SKILLS

Languages	Python, Java, C, C++, Arduino, MATLAB, Javascript, Dafny, Coq, ARM Assembly
Tech & Frameworks	PyTorch, PyG, ROS/2, Numpy, Pandas, TensorFlow/tflite, sklearn, JAX, CoreML, Gymnasium, Sagemaker, OpenCV, AWS, ONNX, Caffe, Anaconda, Jupyter, Docker, CUDA, Matplotlib, Google NNAPI, PyBullet, MuJoCo

INDUSTRY RESEARCH EXPERIENCE

Applied Science Intern, Amazon Robotics. — May 2024 - Jan 2025

- Pre-trained and fine-tuned foundation models under address severe data imbalance for robotic stowing of unknown objects into cluttered bins under occlusion.
- Led research contributing to a non-provisional patent application.

ACADEMIC RESEARCH EXPERIENCE

Research Assistant, Purdue University, IDEAS Lab — Aug 2022 - Present

- Working on representation learning for robotics with Prof. Aniket Bera. Current projects involve multi-modal representations for long-horizon recovery aware manipulation and learning behavior dynamics for multi-agent planning.
- Previously worked on vision-language models for zero-shot grasping (IROS 26), rearrangement manipulation planning using structural concept learning (CoRL 23), and language models for constraint-based interactive navigation (ECCV 26).

Research Assistant, University of Massachusetts — Aug 2021 - May 2022

- Worked on persistent movement tracking for computer vision systems through the lens of perspective tracking with Prof. Erik Learned-Miller at *UMass Computer Vision Lab*.
- Honors Thesis (Improving Object Recognition and Tracking through Alignment and Optical Flow) available at UMass SCAU.

Robotics Institute Summer Scholar, Carnegie Mellon University — Jun 2021 - Sep 2021

- Worked on modeling "trust" between human operators and robots working towards a common goal from an interpretability and performance perspective with Prof. Katia Sycara at the *Advanced Agent-Robotics Technology Lab*.
- Inverse Reinforcement Learning and Bayesian Probabilistic modeling for trust inference in a multi-agent system. Project presented and published at RISS 2021.

Research Intern, Washington University — May 2020 - May 2021

- Worked on combating deepfakes by establishing image provenance which is robust to black-box adversarial attacks and permissive of benign transformations, with Prof. Yevgeniy Vorobeychik. Project published and presented at WACV 2022.
- Implemented a server pipeline to demonstrate proof of concept which ensures and quantifies image authenticity using its provenance. Derived and used homography estimation to detect affine transformation between pair of images using face correspondence.

INDUSTRY EXPERIENCE

Engineering Intern, Neurala Inc. — May 2019 - Aug 2019

- Implemented (from the ground up) a Android interface for an ML framework, which allows for training, inference, combining pre-trained neural nets and more – either locally (hardware accelerated) or seamlessly on a remote server machine.
- Created a prototype/demo for showcasing inference for cashier-less checkout using a camera.

INVITED TALKS

- Concept Learning for Interpretable and Efficient Robotic Agents, University of Washington (Robotics Colloquium), Nov 2024.
- Transfer learning for Robotics, Amazon Robotics (Knowledge Transfer Series), Oct 2024.
- Learning Multi-Agent Dynamics for Herding, Purdue University (Computer Vision and Friends), Oct 2023.

AWARDS AND HONORS

Bay State Fellowship, Commonwealth Honors College Recognition, UMass Departmental Statistics Award, Outstanding TA Award (2 years), Former Faculty Search Representative, Former Vice President for ACM Machine Learning Club, Dean's List (all semesters), MIT Blueprint Creativity Award, FTC World Championship, FTC Design award, FTC Innovation Award